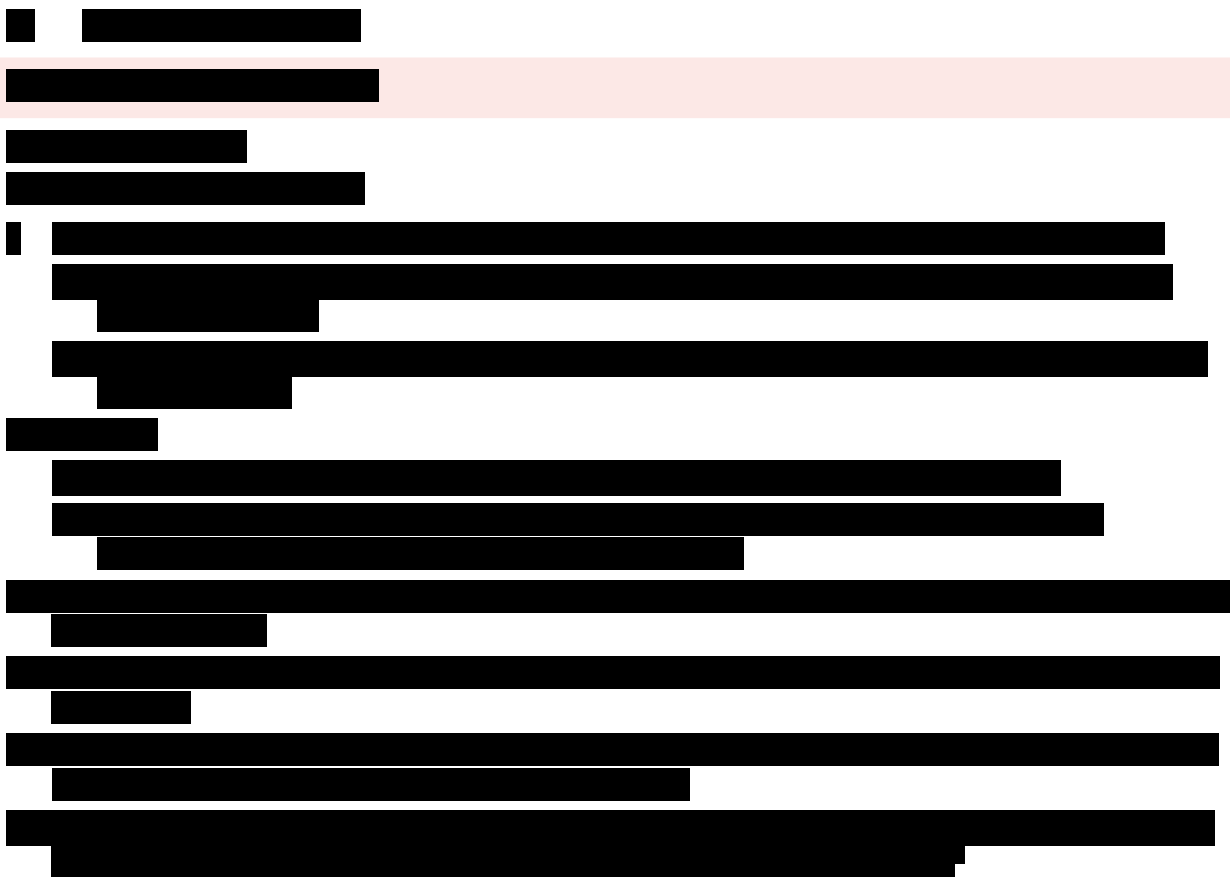




18 Carboxylic acids and derivatives

18.1 Carboxylic acids

Learning outcomes

Candidates should be able to:

- 1 recall the reactions by which carboxylic acids can be produced:
 - (a) oxidation of primary alcohols and aldehydes with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and refluxing
 - (b) hydrolysis of nitriles with dilute acid or dilute alkali followed by acidification
 - (c) hydrolysis of esters with dilute acid or dilute alkali and heat followed by acidification
- 2 describe:
 - (a) the redox reaction with reactive metals to produce a salt and $\text{H}_2(\text{g})$
 - (b) the neutralisation reaction with alkalis to produce a salt and $\text{H}_2\text{O}(\text{l})$
 - (c) the acid–base reaction with carbonates to produce a salt and $\text{H}_2\text{O}(\text{l})$ and $\text{CO}_2(\text{g})$
 - (d) esterification with alcohols with concentrated H_2SO_4 as catalyst
 - (e) reduction by LiAlH_4 to form a primary alcohol

18.2 Esters

Learning outcomes

Candidates should be able to:

- 1 recall the reaction (reagents and conditions) by which esters can be produced:
 - (a) the condensation reaction between an alcohol and a carboxylic acid with concentrated H_2SO_4 as catalyst
- 2 describe the hydrolysis of esters by dilute acid and by dilute alkali and heat

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33 Carboxylic acids and derivatives

33.1 Carboxylic acids

Learning outcomes

Candidates should be able to:

- recall the reaction by which benzoic acid can be produced:
 - reaction of an alkylbenzene with hot alkaline KMnO_4 and then dilute acid, exemplified by methylbenzene
- describe the reaction of carboxylic acids with PCl_3 and heat, PCl_5 or SOCl_2 to form acyl chlorides
- recognise that some carboxylic acids can be further oxidised:
 - the oxidation of methanoic acid, HCOOH , with Fehling's reagent or Tollens' reagent or acidified KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to carbon dioxide and water
 - the oxidation of ethanedioic acid, HOOC-COOH , with warm acidified KMnO_4 to carbon dioxide
- describe and explain the relative acidities of carboxylic acids, phenols and alcohols
- describe and explain the relative acidities of chlorine-substituted carboxylic acids

33.2 Esters

Learning outcomes

Candidates should be able to:

- 1 recall the reaction by which esters can be produced:
 - (a) reaction of alcohols with acyl chlorides using the formation of ethyl ethanoate and phenyl benzoate as examples

33.3 Acyl chlorides

Learning outcomes

Candidates should be able to:

- 1 recall the reactions (reagents and conditions) by which acyl chlorides can be produced:
 - (a) reaction of carboxylic acids with PCl_3 and heat, PCl_5 or SOCl_2
- 2 describe the following reactions of acyl chlorides:
 - (a) hydrolysis on addition of water at room temperature to give the carboxylic acid and HCl
 - (b) reaction with an alcohol at room temperature to produce an ester and HCl
 - (c) reaction with phenol at room temperature to produce an ester and HCl
 - (d) reaction with ammonia at room temperature to produce an amide and HCl
 - (e) reaction with a primary or secondary amine at room temperature to produce an amide and HCl
- 3 describe the addition–elimination mechanism of acyl chlorides in reactions in 33.3.2(a)–(e)
- 4 explain the relative ease of hydrolysis of acyl chlorides, alkyl chlorides and halogenoarenes (aryl chlorides)

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